AI FOR FOUNDATIONAL LEARNING HACKATHON 2026

Samajh

Solution Report

Features, app flow and design of a teacher-first reading check for Class 1 to 5 (Hindi and Marathi)

Samajh means understanding.

CHALLENGE AREA	Learning-Level Visibility
TEAM	Runtime Terrors
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INSTITUTION	Vishwakarma Institute of Technology, Pune
VERSION	1.0 (draft) · 17 October 2026

1. Summary

Samajh is an Android app that a teacher uses to check how a child in Class 1 to 5 reads, what the child understands and how the child writes. It then tells the teacher what to teach next. It addresses the Learning-Level Visibility challenge: a speed score alone can rank a child who understood almost nothing above a child who understood the story but decodes slowly.

This report does three things. It organises the team's 23 features into one flow that a teacher can follow. It adds 10 features that strengthen the solution: 6 core features and 4 that only work with internet. And it specifies how the app should look and behave so that it stays simple for a teacher who is checking children in a busy classroom.

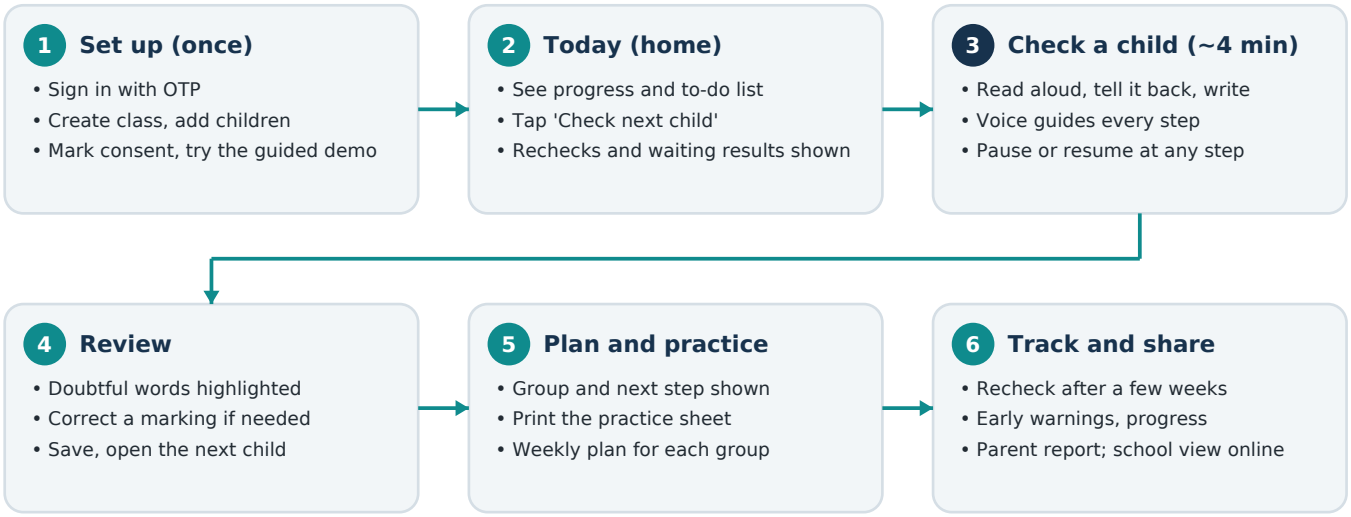
23	10	4	~4 min
features from the team list, all kept	new features (6 core, 4 internet-only)	tabs in the whole app	to check one child

Design principles

Principle	What it means in the app
The teacher is the only user	The teacher operates the app. The child only reads, talks and writes. There are no child-facing screens, accounts or games.
One screen, one job	The home screen has one main button. During a check, each screen shows one instruction with one large button.
Offline first, better online	Reading is scored on the phone. Anything that needs the cloud is saved and finished later, so the teacher never waits for a network.
Explained and correctable	Every result is shown in plain words, with the reason, and the teacher can correct a marking before it is saved.
Every result ends in an action	Each child gets a group, a next step and a practice sheet. The class gets a weekly plan. No result is only a number.

2. How the app is organised

The app follows the teacher's real day: set up once, open the app, check a child, review, hand out practice, and track progress. The six stages below are the same stages used to group the features in section 4.



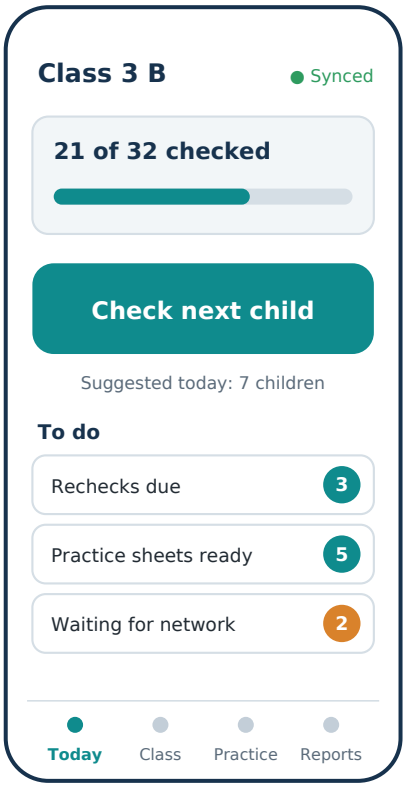
Navigation: four tabs only

Tab	Purpose	What is inside
Today	Home. What to do right now.	Progress for the class, 'Check next child' button, short to-do list (rechecks due, sheets ready, results waiting for network).
Class	Children and groups.	Roster with status of each child, four reading groups, common mistakes, early warnings, weekly plan. Tapping a child opens the child result.
Practice	Material for tomorrow.	Practice sheets to print, weekly activities with tick-off, links to DIKSHA, NCERT and state resources (online).
Reports	Progress and sharing.	Progress over time, parent reports and voice notes, school-level view for the head teacher (online), PDF export.

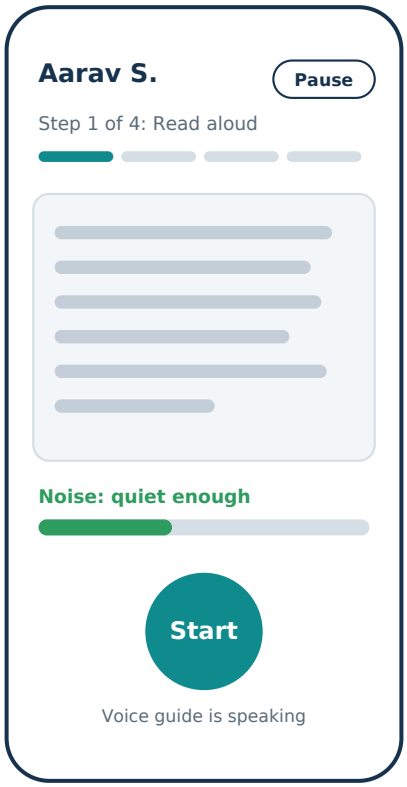
Settings (language, PIN, backup, sync status) sit behind the profile icon at the top right, so they do not crowd the four tabs. The head teacher is a teacher with one extra view inside Reports. It is the same app, not a second app.

3. Key screens

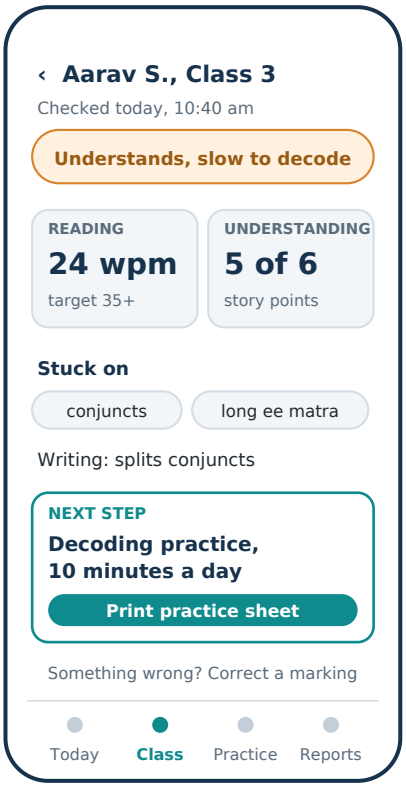
Three screens carry most of the experience. They are shown as simple wireframes, not final visual design.



1 Today (home)



2 Check a child: read aloud



3 Child result

Rules that keep the app simple

- **Home has one main action.** 'Check next child' is the largest element. The rest of the screen is a short to-do list of at most three lines.
- **Plain words first, numbers second.** The result screen leads with the group name and one next step. Speed, pause counts and confidence sit behind a 'details' tap.
- **Same words everywhere.** 'Group', 'practice sheet' and 'recheck' mean the same thing on every screen.
- **Large targets, voice guidance.** Buttons are large for use while standing, and the voice speaks each instruction in the child's language.
- **The network is never a blocker.** If something needs internet, the app says 'Saved, will finish when online' and moves on.
- **Honest about doubt.** When the app is unsure (noise, blurry photo), it says so and asks the teacher to confirm instead of guessing.

4. Feature plan

All 33 features are listed in the order a teacher meets them. **Source** shows whether a feature came from the team's list or is new in this report. **Works** shows whether it needs internet. **Phase**: P1 is the core build for the demo, P2 is added during the pilot window if time allows, P3 is the roadmap after the hackathon.

A. Set up and home

Goal: the teacher is ready to check the first child within five minutes.

#	Feature	What it does	Source	Works	Phase
A1	Teacher sign-in (OTP)	Sign in once with mobile number and OTP. After that a 4-digit PIN unlocks the app, even offline. Each teacher sees only their own classes.	Team list	Both	P2
A2	Class setup and roster	Create a class (grade, section, school language) and add children by typing or importing a list. Each child has a home language and a consent mark.	**New**	Offline	P1
A3	Privacy and child safety	A child can be checked only after parental consent is marked. Audio stays on the phone, photos are deleted after processing, names are blurred, data is encrypted.	Team list	Both	P1
A4	Multilingual support	Screens and voice instructions in Hindi, Marathi and English. Retells may mix Hindi and Marathi. Other languages use a 30-second teacher checklist.	Team list	Offline	P1
A5	Today screen (home)	One screen with progress ('21 of 32 checked'), one button 'Check next child' and a short to-do list: rechecks due, sheets ready, results waiting for network. Suggests how many children to check today.	**New**	Offline	P1
A6	Guided first check	A 2-minute practice run with a sample child, so a new teacher learns the four steps before using the app on a real child.	**New**	Offline	P2

B. Check a child

Goal: finish one child in about 4 minutes, with the teacher doing as little as possible.

#	Feature	What it does	Source	Works	Phase
B1	Adaptive passage selection	Chooses a passage at the child's level. The first check starts slightly below grade, later checks start where the last one ended. An easier passage is offered if the child struggles early.	Team list	Offline	P2
B2	Voice-guided instructions	The app speaks each instruction in the child's language, so the teacher can stay with the child instead of reading prompts.	Team list	Offline	P1
B3	Classroom noise detection	Checks background noise before recording. If it is too loud, the app asks the teacher to move somewhere quieter. If they continue, the result is marked low-confidence.	Team list	Offline	P1
B4	Read-aloud assessment	The child reads a short passage for about a minute. The phone marks skipped, misread and self-corrected words, speed and pauses.	Team list	Offline	P1
B5	Retell (comprehension check)	The child tells the story back in Hindi, Marathi or a mix. The retell is scored against the story's main points (for example 5 of 6).	Team list	Both	P1
B6	Dictation and handwriting	The app reads out five words taken from the child's reading mistakes. The child writes them in a notebook and the teacher takes one photo. The handwriting is read and checked against the known words.	Team list	Both	P1

#	Feature	What it does	Source	Works	Phase
B7	Listening check	Only for a child who struggles with both reading and understanding: the app reads a short passage aloud and the child retells it. This separates a decoding problem from a language problem.	**New**	Both	P2
B8	Pause and resume	A check can be paused at any step and resumed later without starting again.	Team list	Offline	P1
B9	One-tap next student	After saving, one tap opens the next unchecked child.	Team list	Offline	P1

C. Review and results

Goal: the teacher trusts the result and knows what it means.

#	Feature	What it does	Source	Works	Phase
C1	Teacher review and correction	Before saving, words the app was unsure about are highlighted. The teacher can correct a marking, accept a regional pronunciation or retake a step.	**New**	Offline	P1
C2	Mistake mapping	Each error is broken into consonant, matra, conjunct, halant or anusvara, so the app can say 'long ee matra' instead of 'word wrong'.	Team list	Offline	P1
C3	Automatic reading group	Places each child in one of four groups using decoding (speed and accuracy) and understanding, shown in plain words with one next step.	Team list	Both	P1

D. Plan and practice

Goal: tomorrow's lesson is ready without the teacher preparing material.

#	Feature	What it does	Source	Works	Phase
D1	Personalized practice worksheets	A one-page sheet per child with a short passage and a writing task built around the sounds they are stuck on, set in familiar places such as a market or a village fair.	Team list	Both	P1
D2	Practice material validation	A rule-based checker confirms every word uses only letters the child has already learned. A passage that fails is rewritten before printing.	Team list	Both	P1
D3	Weekly class plan	Children with the same gap are grouped, each group gets a 10-minute activity, and the week fits on one printed page. The teacher ticks activities as done.	**New**	Offline	P2
D4	Educational resource links	Links each learning gap to matching DIKSHA, NCERT and state resources, where public links exist.	Team list	**Online**	P3
D5	Progress stars	Teacher-awarded stars for milestones (moving up a group, reaching the benchmark), printed on the child's sheet and a class chart. No game screens for children.	Team list	Offline	P3

E. Track the class

Goal: see who is improving and who is falling behind, in time to act.

#	Feature	What it does	Source	Works	Phase
E1	Teacher dashboard	Class view with reading groups, the most common mistakes (for example, conjuncts: 9 children) and the children who need extra support.	Team list	Offline	P1
E2	Progress tracking	Rechecks after a few weeks compare reading, understanding and writing with earlier results, so the teacher can see whether practice helped.	Team list	Offline	P2

#	Feature	What it does	Source	Works	Phase
E3	Early-warning flags	Flags children unlikely to reach the grade benchmark by March at their current pace, while there is still time to help.	Team list	Offline	P2

F. Share and cloud

Goal: use the internet when it is there, without ever making the teacher wait for it.

#	Feature	What it does	Source	Works	Phase
F1	Catch-up queue	When offline, retell and notebook-photo checks are saved on the phone and processed automatically when the network returns. Each child shows 'Saved' or 'Ready'.	**New**	**Online**	P1
F2	Cloud backup and restore	Results are backed up, so a lost or replaced phone does not lose a term's work. Signing in on a new phone restores the class.	**New**	**Online**	P2
F3	Fresh passages and updates	When online, the app downloads new validated passages, practice sheets and model updates, so the offline bank does not go stale.	**New**	**Online**	P3
F4	Regional pronunciation library	Pronunciations that teachers accept are collected as text (no audio, no names) and shared back as updates, so speech marking improves for each region.	**New**	**Online**	P3
F5	Parent report and home activity	A one-page report in simple words with one home activity. The teacher approves it before it is shared.	Team list	Both	P2
F6	Voice-note parent report	A short Hindi or Marathi voice version of the report for parents who do not read easily, sent by WhatsApp or SMS.	Team list	**Online**	P3
F7	School-level dashboard	Head teachers compare reading progress across classes. Only summary figures are shown, never individual audio or photos. Exportable as PDF.	Team list	**Online**	P3

5. What only works with internet

Samajh stays fully usable offline, but it should get more useful when a connection exists. Four new features are built around that idea. Three features from the team's list also depend on internet: resource links (D4), voice-note sending (F6) and the school-level dashboard (F7).

Feature	Why it needs internet	Benefit to teacher and child	If offline
F1 Catch-up queue	Retell scoring, handwriting reading and fresh practice sheets use cloud AI.	The teacher never repeats or waits on a check. Results arrive when the phone reconnects, for example on school Wi-Fi or at home.	The check is still completed and saved. The reading score shows at once and the rest is marked 'waiting'.
F2 Cloud backup and restore	Needs cloud storage.	A lost or replaced phone does not lose months of results. Progress tracking can run across the whole school year.	An encrypted copy stays on the phone. A reminder to back up appears when online.
F3 Fresh passages and updates	Downloads new content and model updates.	Children do not keep reading the same passages. New language packs and fixes reach every phone without reinstalling.	The app uses the passages and sheets already downloaded.
F4 Regional pronunciation library	Collects and redistributes teacher corrections.	Fewer false mistakes for children who speak a regional variant. Marking improves for each region as teachers accept pronunciations.	The teacher's own corrections still apply on that phone.

Design rule: no feature may block the teacher because internet is missing. Anything that needs the network is queued on the phone, shown as 'waiting', and finished automatically later.

Offline and online behaviour side by side

Area	Without internet	With internet
Sign-in	4-digit PIN after the first sign-in.	OTP on first sign-in and on a new phone.
Read-aloud scoring	Full scoring on the phone: speed, accuracy, pauses, errors.	Same. Results are also backed up.
Retell	Recorded and transcribed on the phone, then placed in the catch-up queue.	Scored against the story's main points; result appears automatically.
Handwriting	Photo taken and stored encrypted. Teacher sees 'waiting'.	Handwriting read and compared with the five expected words. Photo deleted afterwards.
Reading group	Reading result shows at once. Group is confirmed once the retell is scored.	Group assigned automatically.
Practice sheet	Chosen from a stored bank of already validated sheets that match the child's gaps.	A fresh passage is written for the child, then checked by the rule-based validator.
Dashboard and warnings	Built from the results on the phone.	Same, using the backed-up history.
Parent report	The one-page report can be prepared.	Voice note created and sent by WhatsApp or SMS after the teacher approves.
Backup	Encrypted local copy only.	Cloud backup and restore.
Resources and school view	Not available.	DIKSHA, NCERT and state links. School-level dashboard for the head teacher.
Updates	Uses downloaded passages and models.	New passages, language packs and pronunciation updates download.

6. Privacy and child safety

The app handles recordings and photos of young children, so privacy is part of the design and not an add-on. These rules apply across all features.

Area	Rule
Consent	A child cannot be checked until a parent's consent is marked on the roster. The app records the mark; how consent is collected follows each school's process.
Audio	Recorded and transcribed on the phone. Audio is not uploaded. Only the text of a retell, without the child's name, is sent for scoring.
Photos	Names are blurred before upload. The photo is deleted once the words are read. Only the word-level result is kept.
Storage	Results are encrypted on the phone, in transit and in the cloud.
Access	OTP sign-in plus a PIN. A teacher sees only their own classes. A head teacher sees summary figures for the school, not individual audio or photos.
Sharing with parents	Nothing is sent until the teacher approves it. A report contains only that child's own result and one home activity.
Deletion	A teacher can delete a child's record, and it is removed from the backup too.

A legal review against India's Digital Personal Data Protection Act, 2023, which has specific rules for children's data, should be done before any real-school rollout.

7. Build priority and success measures

Phase	When	Features
P1 Core (18)	Build and demo, up to 1 Nov	Class setup and roster, Privacy and child safety, Multilingual support, Today screen (home), Voice-guided instructions, Classroom noise detection, Read-aloud assessment, Retell (comprehension check), Dictation and handwriting, Pause and resume, One-tap next student, Teacher review and correction, Mistake mapping, Automatic reading group, Personalized practice worksheets, Practice material validation, Teacher dashboard, Catch-up queue
P2 Add if time allows (9)	Pilot window, 10 to 18 Nov	Teacher sign-in (OTP), Guided first check, Adaptive passage selection, Listening check, Weekly class plan, Progress tracking, Early-warning flags, Cloud backup and restore, Parent report and home activity
P3 Roadmap (6)	After the hackathon	Educational resource links, Progress stars, Fresh passages and updates, Regional pronunciation library, Voice-note parent report, School-level dashboard

P1 is a large set for a short build. If time runs short, the safest items to cut from P1 are the Today screen suggestions and the full teacher review screen, keeping only a simple 'correct this word' option. The reading, retell, handwriting, groups and practice sheet flow should not be cut.

How we will know it works

Measure	Target	Source
Agreement with a teacher's marking	At least 90%	Idea submission
Time to check one child	Under 5 minutes	Idea submission
Practice passages that pass the decodability check	100%	Idea submission
Audio leaving the phone	None	Idea submission
Runs on a phone with 2 to 3 GB RAM	Yes, tested in pilot	Idea submission
A new teacher completes a first check without help	Within 10 minutes of installing (to be validated)	Proposed
Checks lost when the phone is offline	None	Proposed

Open questions to confirm

- **DIKSHA, NCERT and state resources:** confirm what public links or APIs exist before promising deep integration.
- **Voice notes:** choose a Hindi and Marathi text-to-speech option (for example from AI4Bharat or Bhashini) and test its quality.
- **Consent:** confirm how schools collect parental consent so the roster mark matches real practice.
- **School accounts:** decide how a head teacher is linked to the classes of other teachers.
- **Legal review:** data-protection check for children's data before any pilot with real children.